DEVANSH DUTT PATHAK

Embedded Systems Engineer | Firmware Development | Embedded Hardware | UAV Systems | Industrial IoT | Pan India

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with 1+ year of experience in embedded firmware and hardware development across UAV, Industrial IoT, and smart electronic products. Skilled in Embedded C/C++, STM32, ESP32, FreeRTOS, communication protocols (UART, SPI, I2C, CAN, MQTT), sensor interfacing, hardware debugging, and system integration. Strong foundation in developing reliable, real-time embedded solutions with a passion for innovation and continuous learning.

EXPERIENCE

IG Defence — Noida **May 2026 – Present**

Future Development (R&D) Intern (Paid)

- Integrated Pixhawk flight controller with UAV engine and peripheral subsystems.
- Tested Kamikaze Drone Engines (560 CC) & Vtol Engines (232 CC) for various applications & connected their peripheries
- Developed CAN Bus based engine telemetry acquisition and monitoring systems.
- Worked with ArduPilot and PX4 flight stacks for UAV configuration and testing.
- Implemented real-time telemetry pipelines for engine and flight parameters.
- Supported UAV system integration, testing, and validation activities.

ENLOG — Gurugram **Feb 2026 – May 2026**

Firmware Engineering Intern (Paid)

- Contributed to the development of the EnMate (ESP32-C3 based) smart energy metering product by implementing OTA firmware updates, MQTT-based remote monitoring, remote device control, and calibration tools.
- Assisted in power supply design and created hardware schematics using KiCad for embedded IoT products.
- Developed firmware using ESP-IDF for the ESP32-C3, including bare-metal I2C driver development and sensor data acquisition for the ENS-360 environmental sensor.
- Contributed to the R&D of the En-Sense environmental monitoring product, integrating sensors and validating firmware functionality.
- Diagnosed and resolved PCB, power integrity, and communication issues using oscilloscopes and multimeters, and performed PCB rework and soldering.

IIT Delhi – AIA Foundation for Smart Manufacturing — New Delhi **Jun 2025 – Dec 2025**

Industrial IoT & Automation Intern (Paid)

- Developed firmware for retrofitting conventional industrial machines with IoT capabilities.
- Designed and integrated sensor interfaces for vibration, spindle speed and temperature monitoring.
- Worked with Raspberry Pi and Arduino Mega for industrial monitoring applications.
- Implemented OPC-UA communication stack via KepServerEX; built real-time data pipelines from edge firmware to ThingWorx cloud dashboards
- Delivered Industrial IoT training to engineers to Automobile Engineers; mentored IITD students on firmware design & hardware interfacing

JKSD Infotech Pvt. Ltd. — Noida **May 2024 – Aug 2024**

Embedded Systems Intern

- Developed embedded applications using ESP32 and Arduino for sensor-based IoT applications.
- Integrated sensors using I2C, SPI and UART communication protocols. • Performed PCB soldering, hardware assembly and circuit debugging
- Tested embedded systems using multimeter and oscilloscope. • Assisted in electronic prototyping and hardware validation

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PROJECTS

UAV Telemetry & Engine Monitoring System

- Developed CAN Bus-based telemetry acquisition for real-time UAV engine monitoring and data logging.
- Worked on 560 cc and 232 cc UAV gasoline engines, integrating engine telemetry with embedded monitoring systems.
- Integrated Pixhawk Flight Controller with telemetry modules and peripheral subsystems for synchronized flight and engine data acquisition.
- Implemented cloud-based visualization of engine health and flight parameters for remote monitoring and diagnostics.
- Worked with MAVLink communication, Mission Planner, and telemetry protocols for configuration, testing, and data analysis.
- Assisted in system integration, troubleshooting, and validation of engine electronics and telemetry communication during ground testing.

En-Mate Energy Metering System

- Developed a custom calibration dashboard for configuring and calibrating smart energy meters during production and field deployment.
- Implemented MQTT-based real-time energy monitoring, enabling remote acquisition of voltage, current, power, energy, and other metering parameters.
- Integrated OTA (Over-the-Air) firmware update functionality for remote firmware upgrades without physical access to the device.
- Assisted in implementing remote device control features, allowing configuration and management through cloud connectivity.
- Participated in firmware debugging, hardware validation, and testing to ensure reliable field performance.

ENS-360 Air Quality Monitor — Bare-Metal I2C Firmware Driver

- Developed a bare-metal I2C driver for interfacing the ENS-360 air quality sensor with the ESP32 microcontroller.
- Integrated the sensor into the embedded firmware for real-time acquisition of environmental parameters.
- Designed a local monitoring dashboard to visualize live sensor readings during development and testing.
- Implemented an MQTT gateway for transmitting sensor data to a cloud platform for remote monitoring.
- Performed firmware testing, sensor calibration, and communication debugging to ensure reliable operation.

IIoT Retrofit of Legacy Lathe Machine

- Retrofitted a conventional lathe machine with Industrial IoT capabilities by integrating multiple sensors for machine condition monitoring.
- Developed embedded firmware for acquiring spindle RPM, vibration, feed position, temperature, and electrical energy consumption.
- Integrated sensors including rotary encoder, accelerometer, linear potentiometers, and an RS485-based energy meter.
- Established an OPC-UA communication pipeline using Raspberry Pi and KepServerEX to transfer machine data to ThingWorx dashboards.
- Built a real-time monitoring system for machine health, production parameters, and energy consumption as part of an Industry 4.0 implementation.
- Performed hardware integration, sensor calibration, communication debugging, and end-to-end system validation.

TECHNICAL SKILLS

Firmware / Languages: Embedded C (primary), C++, Python (scripting/automation) ‘
AUTOSAR: AUTOSAR Layered Architecture, MCAL, Basic Software (BSW), Runtime Environment (RTE), Application Layer, ECU Abstraction Layer, Complex Device Drivers (CDD)
UAV Technologies: Pixhawk, ArduPilot, PX4, Mission Planner, CAN Bus Telemetry
Frameworks & RTOS: ESP-IDF, CubeMX, MQTTx, Arduino IDE, FreeRTOS, bare-metal MCU programming
Microcontrollers: ESP32, Arduino Uno/Mega , STM32 (ARM Cortex-M), Raspberry Pi, Pixhawk
Protocols: I2C, SPI, UART, CAN, MQTT, OPC-UA, LoRa, BLE
Hardware Debug: PCB fault diagnosis, multimeter, oscilloscope, soldering & rework, signal integrity analysis
PCB & Tools: KiCad (schematic + layout), LTSpice, KepServerEX, ThingWorx, TIA Portal, PLC (Siemens, Rockwell)

EDUCATION

B. Tech – ECE | A.P.J. Abdul Kalam Technical University, Lucknow (Formerly Uttar Pradesh Technical University, Lucknow) 2022 – 2026

CERTIFICATIONS

Analog ICs & VLSI Design – NIT Delhi (2025)

Robotics using ROS – ABES Engineering College (2025)